Optimal Forest Management with Carbon Sequestration Credits and Endogenous Fire Risk

Adam J. Daigneault, Mario J. Miranda, and Brent Sohngen

ABSTRACT. *We use a stochastic dynamic profit maximization model to investigate the effects of forest carbon sequestration credits on optimal forest management practices for stands facing wildfire risk. Landowners that periodically thin a stand can increase growth rates and mitigate loss of timber and carbon stocks from wildfire. Results indicate that thinning and shortening rotations are cost-effective strategies to mitigate wildfire risk. Carbon prices cause landowners to delay both their thinning treatments and the final rotation age. Thinning and extending timber rotations are thus a viable climate-change mitigation option even when stands are susceptible to risks of fire.* (JEL Q23, Q54)

I. INTRODUCTION

Forests play a pivotal role in the global carbon cycle. By sequestering carbon from the atmosphere, they provide a cost-effective way to mitigate climate change (see, e.g., Sedjo and Solomon 1989; Stavins 1999; Richards and Stokes 2004). Concern has been raised, however, that a significant amount of the carbon sequestered through afforestation may subsequently be released back to the atmosphere as a result of forest fires. Further, there is emerging evidence that climate change itself could cause forest fires to grow in size and intensity, leading to large losses in timber reserves, habitat, and valuable carbon sinks (Westerling et al. 2006). Reports indicate that over 50 million ha of forestland globally are burned each year, generating upward of 1 billion tC (1 tC = 1,000 kg C) in gross carbon emissions (van der Werf et al. 2006). In the United States alone, fires release more than 80 tC annually, accounting for about 4% of the nation's total emissions (Wiedinmyer and Neff 2007).

Recent evidence, however, indicates that silvicultural management practices, such as mechanically thinning a stand or conducting controlled burns, can significantly reduce future carbon losses associated with uncontrolled wildfires (Stephens and Moghaddas 2005; Graham, McCaffrey, and Jain 2004). Given the high value that forests hold for sequestering carbon, and the potentially large risks associated with emissions from forest fires, it is useful to consider the links between forest management, forest fire risk, and forest carbon sequestration policies. This paper develops a stochastic dynamic optimization model to address this issue.

Mechanical thinning can be an important management tool for a landowner who seeks to maximize both timber and carbon offset benefits. Van Wagner (1977), Curtis et al. (1998), and Stephens and Moghaddas (2005) demonstrate that removing surface fuels and dead and dying trees below the canopy can reduce the frequency and intensity of large crown fires that typically destroy large amounts of merchantable timber. In addition to removing dangerous fuels, thinning also can improve sawtimber yields by eliminating smaller, less produc-

Land Economics • February 2010 • 86 (1): 155–172
ISSN 0023-7639; E-ISSN 1543-8325
© 2010 by the Board of Regents of the University of Wisconsin System

The authors are, respectively, economist, Office of Atmospheric Programs, Climate Change Division, U.S. Environmental Protection Agency; professor, Department of Agricultural, Environmental, and Development Economics, The Ohio State University; and professor, Department of Agricultural, Environmental, and Development Economics, The Ohio State University. The authors would like to thank Steve Newbold, two anonymous reviewers, and the editor for helpful comments. All remaining errors are our own. Any opinions, findings, conclusions, or recommendations expressed in this publication are those of the author and do not necessarily reflect the views of the U.S. Environmental Protection Agency.

tive trees from the stand, thereby limiting competition for space, water, and nutrients (Miller, Clendenen, and Bruce 1988; Haynes 2003). Silvicultural practices implemented to improve timber production and mitigate risk could also increase the quantity of carbon sequestered as growing biomass (Hoen and Solberg 1994), as well as the volume of merchantable timber that can be stored in durable wood products after harvest (Sedjo et al. 1995).

Promoting carbon sequestration in fire prone regions might not be economical because management is inherently costly. Carbon offset credits, however, could encourage different levels of fuel treatment and alter stand management from the typical Faustmann rotation. The degree of the impact will be dependent on credit prices. Developing new incentives to reduce the size and intensity of forest fires through fuel treatments should also ease costly fire suppression efforts that are often provided at the public's expense.

Two distinct strands of the research literature have examined parts of the connections between carbon sequestration credits, forest management, and forest fire risk using a Faustmann-type approach, yet no one has integrated all three parts into one concise analysis. The first strand of literature asks how fire risk affects the optimal rotation period. Early papers by Martell (1980), Routledge (1980), Reed (1984, 1987), and Reed and Errico (1985), explored the relationship between merchantable timber values and fire risk. Research by Reed (1993) and Englin, Boxall, and Hauer (2000) extended the analysis to include nontimber values. Yoder (2004) investigated the effect of fire risk on the optimal rotation of prescribed burns, but not timber rotations, relative to a generic benefits function. Recent work by Amacher, Malik, and Haight (2005) investigated the impact of a single fuel treatment on rotations with generic nontimber values, while Amacher, Malik, and Haight (2006) evaluated the incentives of cost-sharing fuel treatments and fire suppression on managing stands without nontimber values. None of these studies look at the impact of carbon sequestration credits on optimal stand management.

The second strand of literature addresses how incentives to sequester carbon affect the optimal rotation. Most of these applications do not account for uncertainty. Englin and Calloway (1993), van Kooten, Binkley, and Delcourt (1995), and Murray (2000) used deterministic models and found that carbon credits extend the optimal rotation period. Chladná (2007) investigated the role of uncertainty in carbon and timber prices on optimal rotations and found that the harvests vary with the type of price process, the way carbon income is defined, and the selection of discount rates. Pohjola and Valsta (2007) found that carbon credits increase the intensity of thinnings, enhance the total amount of carbon stocks on the stand, and also lengthen the rotation age. Most of the deterministic models suggest that planting trees and holding them longer is optimal when carbon is valued. Holding trees longer, however, exposes stands to greater fire risks. Two studies have examined this question to date (Stainback and Alavalapati 2004; Stollery 2005), and both conclude that that rotation ages decline with increased fire risk, but that the reduction is smaller with higher carbon offset prices. Neither the deterministic studies nor the two stochastic studies account for the possibility that stand managers can engage in risk-mitigating forest management practices.

This paper evaluates the impact of carbon sequestration credits on optimal stand management in a forest that is at risk of a forest fire. In our model, landowners have the ability to mitigate risk and potentially enhance timber yields through periodic mechanical thinning. We develop a stochastic dynamic model that optimizes the thinning age and final rotation age, assuming that the probability of fire is exogenous. While fire probability is exogenous, forest managers control the probable loss associated with a fire through periodic risk mitigation practices, for example,

thinning. The model is parameterized for a typical Douglas-fir stand in the Pacific Northwest of the United States and solved and simulated under alternative fire probabilities, carbon sequestration credit prices, risk mitigation costs, and merchantable timber yield responses to stand management. While our stylized model cannot provide specific insight for all privately owned timberlands in the United States, the management regime that we model suggests fire prone stands are a viable climate-change mitigation option.

This paper contributes to the existing literature in several ways. First, it evaluates the presence of carbon credits on the optimal management regime where fire risk can be mitigated with intermediate fuel treatments. Second, it incorporates silvicultural practices that can increase sawtimber growth and sequester carbon at a faster rate than if there were no intervention. Third, it mirrors typical stand management practices by allowing multiple fuel treatments in a single rotation. Last, its results provide valuable insight for forest managers and policy makers who are considering the contribution of wildfire prone forests in climate-change mitigation policy. In an era when forest fires are becoming more frequent and intense, and there is a growing interest in using forests to mitigate climate change, it is important to develop dynamic models that account for forest management, risk mitigation, timber production, and carbon offset credits in a cohesive framework.

II. ECONOMIC MODEL: OPTIMAL FOREST MANAGEMENT

One would expect landowners with valuable timber to account for fire risk when deciding how to manage their stands. In particular, if the stands are prone to wildfire, managers may consider implementing management practices that reduce the chance that their investment will be destroyed before harvest, subject to the costs of implementing these practices. If landowners also receive benefits from carbon sequestration, they must also account for the lost revenues (or penalties) that may occur if fire damages or destroys their stand. These same landowners, however, must also account for lost revenue when they remove biomass from the stand because it also removes valuable carbon. In this paper, the optimal forest management problem is posed as an infinite-horizon, discrete-time, stochastic dynamic optimization model. Numerical methods are used to solve the model (described below), allowing us to simultaneously determine the optimal management regime and rotation age for a stand that faces catastrophic forest fire risk and benefits from carbon sequestration and timber sales.

Optimal Forest Rotations with the Probability of Fire

Our discrete-time, dynamic stochastic optimal forest management model possesses two continuous state variables and one discrete state variable. The two continuous states are (1) the volume of merchantable timber q at the beginning of the current year measured in cubic meters per hectare, and (2) the level of possible combustible fuel m (coarse woody debris and ladder fuels) in the stand at the beginning of the current year. A discrete state, θ, captures whether or not a forest fire occurs during the current year and is modeled as a random variable, with $\theta = 1$ indicating fire and $\theta = 0$ indicating no fire. For simplicity, the occurrence of a fire is assumed to be an "act of God"[1] (e.g., the result of lightning) that cannot be prevented or controlled by the stand manager, and that occurs with a fixed probability P in any given year.

The forest manager is assumed to take one of three different actions each year, $x = \{1,2,3\}$. If $x = 1$, the manager lets the forest continue to grow for another year without active management; if $x = 2$, the manager

[1] Historical data show that fires in the Pacific Northwest Douglas-fir stands are caused by both humans and lightning; however, in this paper we assume that none of these fires can be prevented and are therefore considered natural.

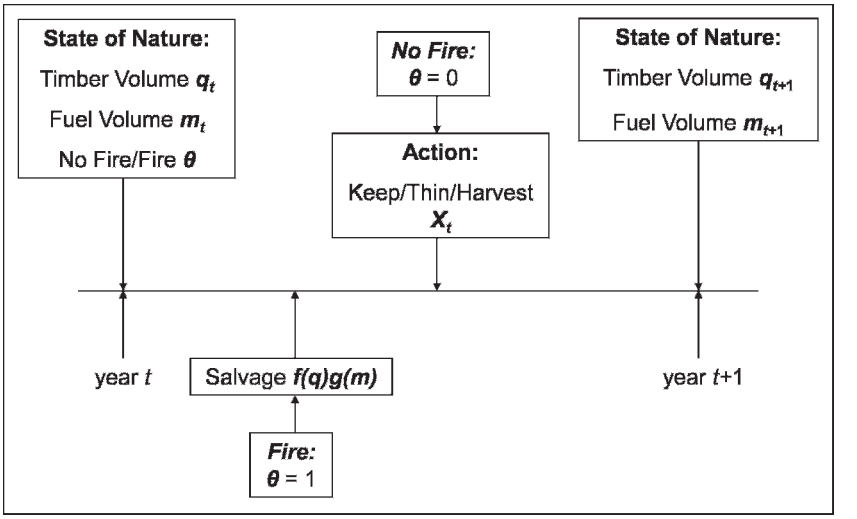

FIGURE 1
TIMING OF STOCHASTIC OPTIMAL ROTATION MODEL

thins the stand to remove excess fuels and improve growing conditions, thereby increasing the growth rate of merchantable sawtimber and decreasing the proportion of timber that is lost if a fire occurs in future periods; or if $x = 3$, the manager clearcuts the stand, sells the merchantable timber, and replants the entire stand. If $\theta = 1$, the forest manager salvages the remaining timber through a clearcut harvest. The timing of the model is shown in Figure 1. The components of the model that allow the manager to make an economically optimal decision are described below.

Fuel Accumulation and Stand Mortality

The stock of fuel, and consequently the change in the stock of fuel in the stand, is determined by management decisions. It is assumed that every time the stand is thinned, clearcut, or salvaged, the amount of fuel is reset to an initial level (m_0).[2] The state of fuel on the stand the following period ($m' = m_{t+1}$) is

$$m' = \begin{cases} k(m) & \text{if } x = 1 \quad \text{(do nothing)} \\ m_0 & \text{otherwise (thin, clearcut,} \\ & \text{salvage harvest).} \end{cases} \qquad [1]$$

In equation [1], m_0 is the initial fuel level, and $k(m)$ is the path of fuel accumulation in the stand over time.

The level of fuel on the stand is also directly related to the proportion of timber that can be destroyed by a fire. In this model, stand mortality, $g(m)$, is formulated as a function of the fuel level, such that $[dg(m)]/dm > 0$ (Omi and Martinson 2002; van Wagner 1977) and is increasing at a decreasing rate. The fire intensity can be reduced through fuel treatments that remove dangerous ladder fuels and coarse woody debris. It is assumed that when the stand is disturbed, either by thinning, clearcut, or salvage harvest after fire, the fuel level and mortality rate are reset to the initial state, such that the proportion of trees destroyed in the event of fire is minimized.

Stand Management and Timber Growth

The level of merchantable timber, q, follows a logistic path typical for biological

[2] In this example, we consider the initial level to be the amount of fuel that minimizes the risk of fire without compromising the growth of the stand. Different management objectives (e.g., increased biodiversity or riparian benefits) might require other fuel levels to achieve maximum benefits.

growth functions and is assumed to increase both as the stand grows naturally over time $f(q)$ and with intermediate management $h(m)$, such that $[df(q)]/dq > 0$, $[dh(m)]/dm > 0$, Empirical studies suggest that thinning from below can increase sawtimber volumes by 15% or more at harvest time than stands that received no silvicultural treatment (Miller, Clendenen, and Bruce 1988; British Columbia Ministry of Forests 1997; Haynes 2003). The timber yield increases from thinning have been reflected in the stand's annual growth function by allowing the manager to control the level of nontimber biomass on the stand, $h(m)$, which is described in more detail below.

The management decision (x), or forest fire ($\theta = 1$) that occurs during the period will impact the levels of fuel and growing stock in the stand. The timber volume for the following period ($q' = q_{t+1}$) is

$$q' = \begin{cases} q_0 & \text{if } \theta = 1 & \text{(salvage clearcut)} \\ f(q)h(m) & \text{if } \theta = 0,\ x = 1 & \text{(do nothing)} \\ f(q)h(m_0) & \text{if } \theta = 0,\ x = 2 & \text{(thin stand)} \\ q_0 & \text{if } \theta = 0,\ x = 3 & \text{(clearcut stand)}, \end{cases} \quad [2]$$

where q_0 is the initial stock planted after a fire and salvage clearcut or planned clearcut at the optimal rotation age, and $f(q)$ is the change in the timber volume during the year (annual growth). The change in quantity that can potentially be logged and converted to timber during the year is also influenced by the level of fuel (both coarse woody debris and ladder fuels) in the stand, $h(m)$. In this model, $f(q)$ and $h(m)$ are assumed separable, such that the fuel levels influence the rate of growth but not the total yield once the stand reaches biological maturity. This follows from the notion that an optimally thinned plantation will increase merchantable sawtimber yields at a faster rate than an unmanaged forest and, hence, further encourage timber managers to invest in intermediate treatments. This paper also conducts a set of sensitivities that relaxes the assumption that increases timber yields to isolate the effect on wildfire risk management.

Merchantable timber represents only a fraction of the total biomass and carbon that accumulates on the stand, as not all of the growing stock can be used for timber products. The level of total growing stock $q_{\text{bio}}(q,m)$ is represented as a function of merchantable sawtimber and the amount of fuel on the stand, such that $[dq_{\text{bio}}(q,m)]/dq > 0$ and $[dq_{\text{bio}}(q,m)]/dm > 0$. Thus, if fuels are removed through intermediate treatment, the merchantable timber yields are increased, but the amount of growing stock and carbon is reduced. This characterization allows us to measure the potential tradeoffs between timber revenues and carbon payments under several fire regimes.

Timber Revenue Benefits

The net revenue received by the forest manager from timber sales or intermediate treatments in any given year, $f_{\text{timber}}(q,m,\theta,x)$, will depend on the volume of merchantable timber q, the volume of combustible fuel m, the incidence of fire θ, and the action taken by the manager x in that year. If a fire occurs ($\theta = 1$), the manager receives the discounted salvage value of the merchantable timber after the fire and replants the stand:

$$f_{\text{timber}}(q,m,\theta,x) = [1 - g(m)]\varsigma p_t q - c_{\text{fire}}. \quad [3]$$

In equation [3], p_t is the deterministic price of the timber (dollars per cubic meter), $0 < \varsigma < 1$ is the fraction of the market price of timber that salvaged logs can be sold for,[3] $g(m)$ is the proportion of timber that is burned and dies if a fire occurs, and c_{fire} is the cost of replanting the stand after the fire (dollars per hectare). Mortality is determined by the amount of fuel in the forest at the time of the fire, and the volume of sawtimber that can be salvaged after a fire is one minus the mortality rate.

If a fire does not occur in a given year ($\theta = 0$), the manager receives revenue that depends on the action taken:

[3] See Baumgartner (1987) and Prestemon and Holmes (2000) for empirical examples.

$$f_{\text{timber}}(q,m,\theta,x)=\begin{cases}-c_{\text{maint}} & \text{if } \theta=0, x=1 \\ & \text{(do nothing)} \\ -c_{\text{thin}} & \text{if } \theta=0, x=2 \\ & \text{(thin stand)} \\ p_{\text{t}}q-c_{\text{rep}} & \text{if } \theta=0, x=3 \\ & \text{(clearcut stand),}\end{cases} \quad [4]$$

where c_{maint}, c_{thin}, and c_{rep} are the respective costs of annually maintaining, thinning, and replanting the stand.

This model assumes that thinning does not produce any commercial timber. Typically, precommercial thinning activities for forest fire management do not produce enough merchantable material to offset the costs of undertaking the thinning activity (Rummer et al. 2003; Barbour et al. 2004). For simplicity, the costs of thinning (c_{thin}) are assumed to be fixed for each intermediate treatment, although we do conduct sensitivity analyses for higher and lower thinning costs.

Carbon Benefits

The model captures carbon benefits associated with having more mature trees and storing carbon in timber products by adding an incentive that captures the carbon stock in trees, $f_{\text{carbon}}(q,m,\theta,x)$. This is achieved by adding a carbon rental price that is established by a given climate-change mitigation policy to the timber benefit function. When forests are used to sequester carbon, the value of the trees is assumed to be a direct function of the carbon embedded in the growing stock or the carbon in the timber harvested and converted to long-lasting wood products. The reward function for forest carbon sequestration is

$$f_{\text{carbon}}(q,m,\theta,x)=\begin{cases}[1-g(m)]\alpha\beta p_{\text{c}}q & \text{if } \theta=1 \\ & \text{(salvage clearcut)} \\ \alpha r p_{\text{c}} q_{\text{bio}} & \text{if } \theta=0,\ x=1,2 \\ & \text{(do nothing, thin)} \\ \alpha\beta p_{\text{c}}q & \text{if } \theta=0,\ x=3 \\ & \text{(clearcut stand).}\end{cases} \quad [5]$$

In equation [5], p_{c} is the price of carbon in dollars per ton of carbon (assumed to be constant in this analysis), α is amount of carbon in tons per cubic meter of biomass, r is the interest rate, and β is the proportion of carbon that is permanently stored in finished wood products after the stand is harvested (i.e., the "pickling rate"). All other variables are defined above.

The carbon rental price approach allows landowners to receive annual payments as long as the stand remains intact. One benefit of a carbon rental payment is that it provides incentives to reduce fuel levels and increase the rotation length, while eliminating the need to pay for carbon lost in a forest fire or other disturbance. Thus, while renting carbon does not penalize current forestland owners by charging them for emissions, removing biomass and carbon through thinning will reduce the amount of the rental payment and could impact the timing and frequency of intermediate treatments. We also assume that landowners receive a payment for converting standing timber to permanent carbon storage, which lets the landowners "trap" the carbon before it can be lost in a fire and could encourage shorter rotations if the pickling rate is high. Therefore, various forest carbon sequestration incentives could have an ambiguous effect on rotation ages and fuel treatments, which is tested in the scenario analysis. Material left on-site is assumed to decay immediately, and landowners do not receive a carbon payment for removing the surface and ladder fuels, as there is no guarantee that it will be used as biomass energy. This approach enables our solutions to represent a lower bound for potential carbon benefits.

Numerical Analysis

The optimal management path can be solved by combining the state variables, the action variables, and the reward and transition functions for both the timber and carbon sequestration markets into a

single Bellman equation:

$$V\{q,m,\theta\} = \max_{x=\{1,2,3\}} \begin{cases} [1-g(m)](p_t\varsigma + \alpha\beta p_c)q - c_{\text{fire}} \ldots \\ \quad + \delta E_{\tilde{\theta}} V(q_0,m_0) & \text{if } \theta = 1 \\ \alpha p_c r q_{\text{bio}} - c_{\text{maint}} + \delta E_{\tilde{\theta}} V(q',m') & \text{if } \theta = 0,\ x = 1 \\ \alpha p_c r q_{\text{bio}} - c_{\text{thin}} + \delta E_{\tilde{\theta}} V(q',m_0) & \text{if } \theta = 0,\ x = 2 \\ [p_t + \alpha\beta p_c]q - c_{\text{rep}} + \delta E_{\tilde{\theta}} V(q_0,m_0) & \text{if } \theta = 0,\ x = 3. \end{cases} \quad [6]$$

$V\{q,m,\theta\}$ is the expected present value of stage returns from applying the derived optimal policy x^* across an infinite amount of stages, starting with a newly planted stand. The maximand is the expected return of the current period from applying the optimal policy decisions plus the expected present value of continuing to apply the optimal policy starting at the next period, which is discounted by $\delta = 1/(1 + r)$. This problem accounts for the stochastic effect of forest fires by incorporating the probability that a random shock, $\tilde{\theta} = \{0,1\}$, will occur equally in each future period.

The endogenous specification and functional equations in this optimal rotation problem do not allow a closed-form solution. Therefore, we approximate a solution to the Bellman equation using the method of collocation (Miranda and Fackler 2002; Judd 1998). This is achieved by approximating the value function approximant as a linear combination of n known basis functions whose coefficients are fixed by requiring that the value function approximant satisfy the Bellman equation, not at all possible states, but rather at the n collocation nodes (Miranda and Fackler 2002). Once the collocation equation has been solved for the n unknown coefficients, residuals from the chosen collocation method are calculated to verify that the approximation errors are minimal across the entire domain of the value function (Miranda and Fackler 2002). For our analysis, cubic spline basis functions were chosen to solve equation [6] in MATLAB 6.1 using the dpsolve routine included in the Miranda and Fackler (2002) COMPECON library. Application of the collocation method is an efficient way to solve numerical problems without imposing additional assumptions on the model or compromising the nonlinear components of the value function. Postoptimality analysis shows that the maximum residuals for both the baseline and scenario analyses were on the order of 10^{-3} times the value of the firm, indicating an acceptably accurate numerical solution to the Bellman equation.

III. EMPIRICAL DATA AND APPLICATION

The model described above is implemented for a representative Douglas-fir (*Pseudotsuga menziesii*) stand in the Pacific Northwest of the United States. Douglas-fir is important as a timber species and, potentially, as a warehouse for carbon sequestration as it can hold over 200 metric tons of aboveground carbon per hectare (Birdsey 1996). Stands in the Pacific Northwest are already susceptible to various levels of forest fire risk depending on the slope, elevation, and weather and climate conditions of the stand. The impact of varying costs, carbon prices, and fire return intervals are tested on optimal management regimes using sensitivity analysis.

The specific functions and parameters used in the base case model are listed in Table 1. Alternative values used in the sensitivity analysis are also listed. The growth and yield functions are for an average site III Douglas-fir with a site-index of 30 m (average tree height at 100 years) were based upon stand tables compiled by McArdle, Meyer, and Bruce (1949, 1961). The effects of thinning on stand growth were estimated based on U.S. Department of Agriculture Forest Service technical reports on silviculture in Douglas-fir plantations (Miller, Clendenen, and Bruce 1988; Curtis et al. 1998) and timber growth projections (Haynes 2003).

The price for timber is given as the average annual real stumpage sale prices (2006 \$/m^3) of Douglas-fir from national

TABLE 1
PARAMETERS AND VALUES USED FOR DOUGLAS-FIR OPTIMAL ROTATION MODEL

Parameter	Symbol	Assumed Form/Value	Alternative(s)
Discount factor	δ	0.95	n/a
Annual timber growth	$h(m)f(q)$	$\left(\frac{2}{1+m}\right)(0.532q - 1 * 10^{-3}q^2 + 1 * 10^{-6}q^3 - 6 * 10^{-11}q^4)$	n/a
Min. timber stock	q_0	4	n/a
Max. timber stock	$q_{\max}$	900	n/a
Forest growing stock	$q_{\text{bio}}(m,q)$	$\left(\frac{5+m}{4}\right)[(900q)^{0.47}] - 26$	n/a
Stumpage price	p_t	60	n/a
Maintenance cost	c_{maint}	10	n/a
Thinning cost	c_{thin}	400	200, 800
Replanting cost	c_{rep}	400	n/a
Salvage cost	c_{fire}	500	n/a
Optimal fuel level	m_0	0.05	n/a
Mortality function	$g(m)$	$m^{0.93}$	n/a
Fuel accumulation function	$k(m)$	$15^{(1+m^{0.93})}$	n/a
Fraction of stumpage price for salvage sales	ς	0.75	n/a
Carbon price	p_c	50	0, 25, 100, 200
Rental rate of carbon	r	$(1 - \delta)/\delta$	n/a
Carbon conversion	α	0.2	n/a
Pickling rate	β	0.4	0.00, 1.00
Probability of fire	$P(\theta)$	$P(\theta = 1) = 0.02$	0.01, 0.04

Note: n/a, not applicable.

forests in Washington and Oregon from 1965 to 2002 (Howard 2003). Carbon prices range from \$0 and \$100 per ton carbon (\$/tC),[4] which are consistent with other studies of climate-change mitigation potential in forestry (Murray 2000; USEPA 2005; Kindermann et al. 2008). Both the timber and carbon prices are assumed to be constant over time. Stand maintenance (c_{maint}) and replanting costs (c_{rep}) are assumed to be \$10/ha/yr and \$400/ha, respectively (Bair and Alig 2006). Thinning costs (c_{thin}) are varied across \$200/ha, \$400/ha, and \$800/ha in the analysis to test the sensitivity of the results to different levels of marketable outputs from thinning as well as stand access costs (Rummer et al. 2003). Postfire replanting costs (c_{fire}) are assumed to be 25% greater than standard replanting costs because of the extra work necessary to prepare the site for regeneration (Reed 1984).

[4] This is equivalent to a range of \$0 to \$27.27 per ton of carbon dioxide (\$/tCO_2e).

The discount factor, δ, is set equal to 0.95 for most scenarios, implying a real rate of interest of approximately 5%. This is consistent with commonly used discount factors in most forest investment research. On average, the probability of a wildfire for many Douglas-fir stands in the Pacific Northwest is once every 50 years (Morrison and Swanson 1990; Agee 1993), however, the actual probability for a site will depend on slope, elevation, and climate. To assess these differences, our scenario analysis models the impacts from the probability of a forest fire once every 25, 50, and 100 years.

The fuel accumulation and mortality rate functions follow research by Brown, Reinhardt, and Kramer (2003), Omi and Martinson (2002), and Smith, Heath, and Jenkins (2003), which show that fuel accumulation and mortality functions generally increase over time at a decreasing rate. The salvage function is based on the potential for a stand-destroying crown fire determined by the current state of the fuel loads. Because there is limited empirical

evidence on the specific shape of this function, we estimated it by conducting multiple runs of the U.S. Department of Agriculture Forest Service's Forest Vegetation Stand (FVS) growth simulator (Dixon 2002) and the FVS Fire and Fuels Extension Model (Reinhardt and Crookston 2003). The FVS model projects stand conditions under a range of thinning alternatives (timber quantity and quality, fuel loads, likely fire severity, and stand mortality under fires with different levels of severity). The FVS simulator was configured for site conditions similar to our representative Douglas-fir stand, then run several times using various fuel loadings and with fires simulated at different points in time to estimate the impact on mortality. The results of several iterations were used to estimate the average fuel accumulation and stand mortality rates. Salvageable timber rates were best represented using a simple but plausible concave function.

IV. RESULTS

A number of different scenarios are developed in order to test the effects of different assumptions about fire frequency, timber yield response, thinning costs, carbon prices, and the pickling of carbon in long-lasting wood products on management intensity and the final rotation age. Both stochastic and deterministic results are presented for each scenario under the blanket assumption that thinning does and does not affect timber yields (Tables 2 and 3). The stochastic results show what happens when simulations are conducted assuming a positive probability of fire. The deterministic results assume that this probability is 0.00 but still allows thinning to increase timber yields.

Yield and Wildfire Mitigation Response to Thinning Results

The baseline scenario in this example assumes that there is no value for carbon sequestration, and the probability of a fire is once every 50 years (P(fire) = 0.02). Landowners reap both wildfire mitigation and timber yield response benefits with thinning. Under these conditions, the optimal rotation length in the stochastic model for Douglas-fir is 51 years. Two thinnings are found to occur optimally at 30 and 39 years after planting (Table 2). A deterministic version of the same model assumes no fire risk, and finds an optimal rotation age of 55 years with one thinning at 38 years, indicating that landowners would still thin the stand to increase timber yields. Thinnings are projected to occur earlier and more frequently when there is a positive probability of fire, and the rotation age is four years younger for the baseline case when there is a probability of fire versus no risk. This suggests that it is optimal for landowners invest in intermediate stand management and reduce their rotation ages to limit the potential losses from fires. Findings for the baseline are consistent with the other fire and optimal rotation literature. The stand's bare land value (BLV)[5] is only slightly higher in the deterministic case ($144/ha)[6] than in the stochastic case ($131/ha).

Scenarios 2 and 3 examine the effect of a change in the probability of a fire, holding all other parameters constant. As expected, a decrease in probability of a fire results in a longer rotation and increases the final harvest; still, there are two thinnings during the rotation with $P = 0.01$. An increase in the probability moves the timing of both

[5] In this analysis, the bare land value is equivalent to the maximum net present value of the timber and carbon benefits from a stand that is harvested over an infinite horizon. The actual price for the land will differ from this present value formula for several reasons, including other nonmarket values, development value, and other factors, all of which are excluded from this analysis.

[6] Baseline BLVs calculated in this paper are relatively low for a Pacific Northwest Douglas-fir stand due to the assumptions about biological growth and merchantable timber yield functions, stand management costs, timber and carbon prices, and the discount rate. The following scenario analysis focuses on parameters that are the most sensitive when optimally managing a stand enrolled in a carbon sequestration program with the risk of wildfire, although modifying other input costs or the yield function would change the BLV estimates.

TABLE 2

Results for Optimal Harvest and Management Model with Timber Yield Response to Thinning

	Baseline	Scenario 2	Scenario 3	Scenario 4	Scenario 5	Scenario 6	Scenario 7	Scenario 8	Scenario 9	Scenario 10	Scenario 11	Scenario 12
Parameters												
Stumpage price ($/m^3)	$60	$60	$60	$60	$60	$60	$60	$60	$60	$60	$60	$60
Discount factor	0.95	0.95	0.95	0.95	0.95	0.95	0.95	0.95	0.95	0.95	0.95	0.95
Thinning cost ($/ha)	$400	$400	$400	$400	$400	$400	$400	$400	$200	$800	$400	$400
Carbon price ($/tC)	$0	$0	$0	$50	$50	$50	$25	$100	$50	$50	$50	$50
Picking rate	0	0	0	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0	1
Probability of fire	0.02	0.01	0.04	0.01	0.02	0.04	0.02	0.02	0.02	0.02	0.02	0.02
Estimates												
Stochastic rotation (years)	51	54	48	58	55	53	52	61	53	57	57	55
Deterministic rotation (years)	55	55	55	64	64	64	58	71	56	64	64	63
Stochastic thinning (years)	30, 39	33, 42	28, 37	37, 46	32, 40	28, 38	31, 39	35, 42	28, 36, 42	41	35, 43	27, 38
Deterministic thinning (years)	38	38	38	0	0	0	38	0	17, 29, 39	0	0	0
Stochastic bare land value ($/ha)	$131	$141	$96	$1,265	$1,261	$1,242	$688	$2,446	$1,348	$1,237	$1,211	$1,353
Deterministic bare land value ($/ha)	$144	$144	$144	$1,299	$1,299	$1,299	$711	$2,543	$1,385	$1,299	$1,253	$1,368

thinnings to earlier in the rotation as landowners act to limit their losses in the event of a fire, but does not necessarily increase the frequency of intermediate treatments. The stand BLV declines in the stochastic model compared to the deterministic solution, but it is still positive.

The implications of carbon prices are introduced in Scenarios 4, 5, and 6, with a price of \$50/tC. The optimal rotation age increases relative to the baseline in each of the scenarios to 58, 55, and 53 years, respectively. Because removing biomass from the stand also reduces carbon payments, there are still only two thinnings per rotation with carbon prices. This means that management does not become more intensive when carbon is valued, although both thinnings and final rotations do occur later. Valuing carbon has a substantial effect on the value of land, as BLV rises substantially, to \$1,265/ha under Scenario 4. If the probability of fire is zero, the optimal harvest age is 64 years and no thinnings are expected to occur. This indicates that leaving the biomass on the stand to store carbon at \$50/tC is more valuable than the yield benefits from thinning it. In all three scenarios where carbon is valued at \$50/tC, BLV calculations for the stochastic and deterministic cases are similar, indicating that active management in a fire prone stand can yield similar expected payoffs if the stand mortality is kept relatively low.

Scenarios 7 and 8 examine the influence of changing the carbon price, but keeping the fire probability consistent with the baseline at 2%. The rotation age increases (decreases) with a higher (lower) carbon price. Both scenarios keep the number of thinnings per rotation at two, but the timings of the thinnings actually increase with carbon price, again pointing to the value of keeping biomass on the stand even with the risk of fire. Thus, in the presence of fire risk, landowners enrolled in a carbon sequestration program will invest more in risk-reducing thinnings throughout the rotation period, and increase rotation lengths regardless of the carbon price and probability of fire.

Altering the net financial impacts of thinning does influence the rotation period and management (see Scenarios 9 and 10). Lower net costs of fuel treatments (e.g., there are many opportunities to convert material to markets nearby) result in more frequent thinnings and higher BLV. Not only are expected losses from fires minimized by frequent fuel removals, but growth is also enhanced. Lower fixed fuel costs also encourage the stand to be managed more intensely (i.e., less time between each thinning), such that timber yields increase and thus shorten the average rotation. This suggests that in regions where there are many opportunities for harvesting pulpwood or biomass material nearby, stands are likely to end up being more highly managed. Of course, if net costs of fuel treatments are higher, fewer thinnings would occur, and stands are under more risk over their lifecycle. If society wishes to reduce the risk of fires through more intensive stand management, it may need to subsidize landowners through treatment costs, but not necessarily provide carbon payments.

Scenarios 11 and 12 assess the effect of changing the pickling rate for carbon that is permanently stored in wood products. In Scenario 11, the pickling rate is set to zero, such that the carbon is assumed to be immediately turned over in wood products and emitted into the atmosphere. When compared to Scenario 5 (P(fire) = 0.02, p_c = \$50), with all other parameters the same except that the pickling rate is 40%, the rotation age increases 2 years, to 57. In Scenario 12, where the pickling rate is 1 (i.e., all timber is permanently stored), the rotation age is the same as in Scenario 5. BLV differs modestly across Scenarios 5, 11, and 12 because of the changes in the payment for permanently stored carbon. Interestingly, the timings of the two thinnings decrease with the pickling rate. These results show that the assumption about how carbon is stored after harvest does affect optimal management decisions.

Only Wildfire Mitigation Response to Thinning Results

As described above, it may be optimistic to assume that landowners gain timber yield when stands are thinned. We thus examine a set of scenarios in which timber yields do not increase when thinning occurs. These scenarios also isolate the effect of using intermediate treatments to reduce wildfire risk (Table 3).

When timber yields do not increase with thinning, it is not surprising that fewer thinnings occur. In fact, when fire probability is 0.01 and carbon is not valued, no thinnings are predicted. A single thinning is predicted for a higher fire probability of 0.02 and 0.04. As in the earlier results, final rotation age is lower with higher fire probability; however, final rotations are longer for the results where there is no yield benefit with thinning. The results are qualitatively similar when carbon is valued at $50/tC, such that there are no thinnings when fire probability is 0.01, there is a single thinning when fire probability is 0.02, and there are two thinnings when fire probability is 0.04. Higher carbon offset credit prices do not necessarily increase the frequency of thinnings, while reducing the costs of intermediate fuel treatment does.

Comparison with Previous Research

Most optimal rotation models assume that the sole adjustment landowners can make in the face of carbon markets or the risk of forest fire is to adjust the rotation age. This is a key choice variable, but there are others as well, including management practices that allow landowners to "fireproof" their forests through intermediate fuel treatments. This section compares the results in this study with those in the earlier papers across several of the scenarios.

We compare our model results to the traditional Faustmann (1849) approach, which ignores fire risk entirely and which does not accommodate thinning, and the Reed (1984) optimal rotation model under risk of wildfire. For the implementation of Reed's model, the fire probability is simply added to the interest rate to calculate the optimal rotation length and BLVs. The exact methods and equations used to calculate the Faustmann and Reed optimal rotation solutions both with and without carbon prices are highly cited in the literature (e.g., van Kooten, Binkley, and Delcourt 1995; Murray 2000). Table 4 shows the results for optimal rotations and BLV for the baseline and some of the alternative scenarios.

The Faustmann and deterministic models, as expected, show no differences across the different fire probabilities. Both show an increase in rotation age and an increase in BLV when carbon prices are incorporated. In the Reed and stochastic models, increasing fire probability reduces the rotation age. Adding carbon prices increases the rotation age in all cases. Of particular interest, however, is the notion that the Reed model suggests that land values are negative when fire probabilities are incorporated and there is no market for carbon sequestration. As fire probability increases, BLV becomes more negative. Under our stochastic model with endogenous fire risk, land values are positive in all cases, even if thinning does not affect timber yields. In fact, the Reed model would indicate that it is not economically efficient at all to invest in timber in the presence of fire risk unless there is a positive price for carbon. Accounting for intermediate treatments results in significantly different outcomes, and could result in planting trees in areas that might not be considered economical under the standard keep or cut rotation models. This ability to conduct thinnings and also to collect carbon payments can have a strong impact on landowner decisions to afforest a fire prone area and enroll it in a forest carbon sequestration program.

Carbon Storage

Our results show that it is feasible to establish and actively manage timber plantations in forests that have fire risk. While it is true that higher fire risk reduces the

TABLE 3

RESULTS FOR OPTIMAL HARVEST AND MANAGEMENT MODEL WITH ONLY WILDFIRE MITIGATION RESPONSE TO THINNING

	Baseline	Scenario 2	Scenario 3	Scenario 4	Scenario 5	Scenario 6	Scenario 7	Scenario 8	Scenario 9	Scenario 10	Scenario 11	Scenario 12
Parameters												
Stumpage price ($/m^3)	$60	$60	$60	$60	$60	$60	$60	$60	$60	$60	$60	$60
Discount factor	0.95	0.95	0.95	0.95	0.95	0.95	0.95	0.95	0.95	0.95	0.95	0.95
Thinning cost ($/ha)	$400	$400	$400	$400	$400	$400	$400	$400	$200	$800	$400	$400
Carbon price ($/tC)	$0	$0	$0	$50	$50	$50	$25	$100	$50	$50	$50	$50
Picking rate	0	0	0	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0	1
Probability of fire	0.02	0.01	0.04	0.01	0.02	0.04	0.02	0.02	0.02	0.02	0.02	0.02
Estimates												
Stochastic rotation (years)	53	54	49	61	60	59	57	64	61	58	61	60
Deterministic rotation (years)	57	57	57	64	64	64	61	71	64	64	64	63
Stochastic thinning (years)	43	0	37	0	43	37, 52	42	47	41, 53	0	48	45
Deterministic thinning (years)	0	0	0	0	0	0	0	0	0	0	0	0
Stochastic bare land value ($/ha)	$25	$80	$11	$1,253	$1,162	$1,090	$585	$2,346	$1,165	$1,232	$1,122	$1,287
Deterministic bare land value ($/ha)	$89	$89	$89	$1,299	$1,299	$1,299	$669	$2,543	$1,299	$1,299	$1,253	$1,368

TABLE 4
COMPARISON OF ENDOGENOUS RISK MODEL WITH STANDARD FAUSTMANN (1849) AND REED (1984) OPTIMAL ROTATION MODELS

Scenario	Carbon Price	*P*(fire)	Faustmann	Deterministic with Yield Response	Deterministic without Yield Response	Reed	Stochastic with Yield Response	Stochastic with Yield Response
Rotation Length (years)								
Baseline	$0	0.02	57	55	57	44	51	53
Scenario 2	$0	0.01	57	55	57	51	54	54
Scenario 3	$0	0.04	57	55	57	41	48	49
Scenario 4	$50	0.01	64	64	64	60	58	61
Scenario 5	$50	0.02	64	64	64	57	55	60
Scenario 6	$50	0.04	64	64	64	42	53	59
Bare Land Value ($/ha)								
Baseline	$0	0.02	$89	$144	$89	($319)	$131	$25
Scenario 2	$0	0.01	$89	$144	$89	($180)	$141	$80
Scenario 3	$0	0.04	$89	$144	$89	($450)	$96	$11
Scenario 4	$50	0.01	$1,299	$1,299	$1,299	$859	$1,265	$1,253
Scenario 5	$50	0.02	$1,299	$1,299	$1,299	$620	$1,261	$1,162
Scenario 6	$50	0.04	$1,299	$1,299	$1,299	$382	$1,242	$1,090

rotation age, regions that face fire risk can still economically sequester carbon by intensifying management and increasing the rotation age with carbon credits. To show this, the following formula is used to calculate the present value of the carbon stock (S_0) over 200 years:

$$S_0 = \int_0^{200} c'(t)e^{-rt}dt, \qquad [7]$$

where $c'(t)$ is the annual change in carbon stock (tons C) in both growing stock and the wood product stock (accounting for turnover), and r is the discount rate of 5%. To account for the carbon turnover in durable wood products, we assume that merchantable sawtimber faces a decay rate of 1% per year (Winjum, Brown, and Schlamadinger 1998). Longer rotations extend the period for which carbon is accumulating as biomass and also delay the time when it is reintroduced to the atmosphere through harvesting and product emissions. Therefore, S_0 is representative of the present value of the carbon stock benefits of bare land allocated to rotational forestry.

Taking the difference between the scenarios with no carbon price and the scenarios with a carbon price estimates the marginal changes in the net present value of carbon stock for 1 ha of forestland. One can then use these differences to generate marginal cost functions across different constant carbon prices. Figure 2 presents these marginal cost functions for the four forest fire probabilities analyzed in this paper for both assumptions about the response of yields to thinning. All other parameters are held at their base case levels. Interestingly, a positive probability of fire leads to higher marginal cost estimates in the case where yields respond to thinning. If there is no response of yield to thinning, marginal costs are similar or slightly lower with a positive probability of fire. This is a surprising result, but recall that the yield responses enhance the sawtimber volume, but not carbon volume. Thus the additional thinnings that occur in the scenarios with the yield response dissipate carbon and raise costs of carbon sequestration.

V. DISCUSSION AND CONCLUSION

This paper develops an optimal rotation model with endogenous risk for a stand with both timber and carbon sequestration benefits in a fire prone region. Silvicultural practices (specifically thinning) that allow risk and growth to be endogenous are introduced in the model. An option for

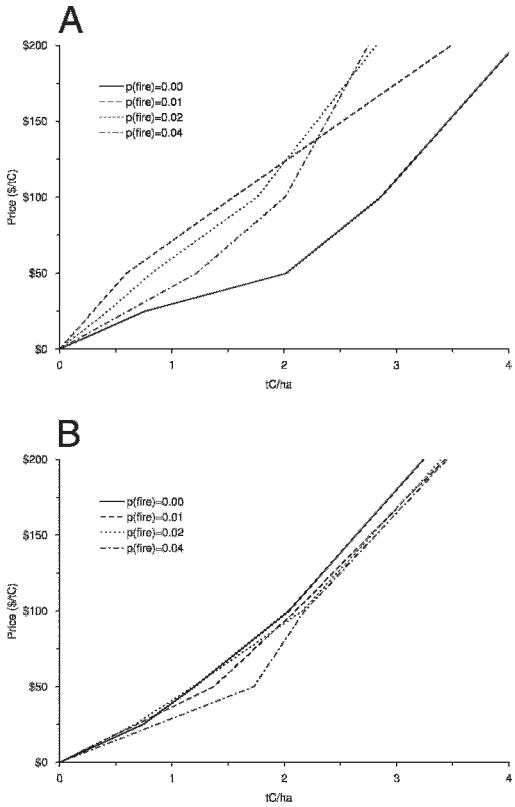

FIGURE 2
MARGINAL COST FOR CARBON SEQUESTRATION WITH (A) AND WITHOUT (B) TIMBER YIELD RESPONSE TO THINNING

multiple intermediate treatments per rotation more accurately portrays the forest manager's ability to optimize both timber and carbon benefits under uncertainty than previous models. Numerical simulations are used to analyze how timberland management and optimal rotations adjust when fire risk is present. Douglas-fir stands in the Pacific Northwest of the United States are used for our numerical example, as they are important as a timber species, have already been established as actively managed plantations, and are a large warehouse for terrestrial carbon. While our stylized model cannot necessarily be applied to all private timberlands, it does provide insight on the potential timber and carbon benefits that thinning can have on many fire prone stands in the United States.

A number of different scenarios are developed in order to test the effects of different assumptions about fire frequency, yield response, thinning costs, carbon prices, and the pickling of carbon in long-lasting wood products on management intensity and the final rotation age. Results indicate that a probability of fire always results in a lower rotation age relative to the deterministic case when there is no risk of fire. Higher carbon prices have a counter-effect though, as the lengths of rotations increase, even in the event that there is a high probability of a fire.

Results also indicate that fuel treatments do not always increase in frequency when carbon prices are high, as landowners are willing to risk the loss of future timber values in order to accrue annual carbon rental payments. In the scenarios where thinning also improved timber yields, results indicate that leaving the biomass on the stand to store carbon at $50/tC could be more valuable than the yield and wildfire mitigation benefits from thinning it. Stands that faced lower thinning costs did increase their timing and frequency of intermediate fuel treatments. This suggests that if society wishes to reduce the risk of fires through more intensive stand management, it may need to subsidize landowners through treatment costs, but not necessarily provide carbon payments.

The marginal costs of carbon sequestration differed significantly with carbon price, fire probability, and yield response to thinning assumption. If managers simply base their strategy on the goal of increasing the forest stock per hectare and extending the rotation age, then forest carbon sequestration policies should still focus on areas with lower probabilities of fire or where forest management and fuel reduction costs are cheapest. High-risk areas that can undergo intermediate fuel treatments should not be immediately ruled out though, as our marginal costs of increasing carbon stock estimates were not always lowest for low or zero probability stands. Results from every scenario presented in this paper suggest that *any* forest fire

mitigation plan should result in net positive carbon sequestration relative to the baseline case where there is no carbon market or fuel treatment.

Costs for fire suppression in the United States have increased significantly over the past decade and exceeded more than a billion dollars in 2007. These costs are not expected to decline in the future, especially if the frequency and intensity of wildfire continues on the upward trend. The results presented in this paper suggest that forest fire management could be more effective in reducing wildfire damage and increasing carbon sequestration in our forests if funds were also spent on prevention (e.g., fuel treatment) programs in addition to suppression.

As the desire to curb global warming continues to rise, the benefits for investing in afforestation efforts and establishing new plantations that can be used for long-term sequestration are expected to increase, as forests are a primary near-term, low-cost solution (USEPA 2005). Furthermore, the expected increase in forest fire frequency over the next century could have significant impacts on carbon stocks (Westerling et al. 2006). The ability to mitigate the loss of timber and carbon in fire prone areas of the western United States further enhances the benefits of investing in plantations and enrolling in carbon sequestration programs. The overall reduction in emissions from forest fires coupled with the increase in carbon sequestration from optimal forest management provide benefits to landowners and society as a whole. Further research is needed to determine the effects of thinning and carbon prices on the optimal management of other stands that might vary in species composition, growth rates, or risk of mortality such as those in other forests in the Pacific Northwest or South. This model also could be extended to research the optimal rotation problem with other nontimber amenities (riparian benefits, water quality, etc.), other risks (hurricanes, invasive species, etc.), and the role of biomass energy markets on large fuel removal programs.

References

Agee, James K. 1993. *Fire Ecology of Pacific Northwest Forests*. Washington, DC: Island Press.

Amacher, Gregory S., Arun S. Malik, and Robert G. Haight. 2005. "Not Getting Burned, the Importance of Fire Prevention in Forest Management." *Land Economics* 81 (2): 284–302.

———. 2006. "Reducing Social Losses from Forest Fires." *Land Economics* 82 (3): 367–83.

Bair, Lucas S., and Ralph J. Alig. 2006. *Regional Cost Information for Private Timberland Conversion and Management*. General Technical Report PNW-GTR-684. Portland, OR: U.S. Department of Agriculture, Forest Service, Pacific Northwest Research Station.

Barbour, R. James, Roger D. Fight, Glenn A. Christensen, Guy L. Pinjuv, and Rao V. Nagubadi. 2004. *Thinning and Prescribed Fire and Projected Trends in Wood Product Potential, Financial Return, and Fire Hazard in Montana*. General Technical Report PNW-GTR-606. Portland, OR: U.S. Department of Agriculture, Forest Service, Pacific Northwest Research Station.

Baumgartner, David C. 1987. "Salvaging Fire-Damaged Timber in Michigan." *Northern Journal of Applied Forestry* 4 (3): 149–52.

Birdsey, Richard A. 1996. "Carbon Storage in United States Forests." In *Forests and Global Change,* Vol. 2: *Opportunities for Improving Forest Management*, ed. R. Neil Sampson and Dwight Hair. Washington, DC: American Forests.

British Columbia Ministry of Forests. 1997. *Forest Level Benefits to Commercial Thinning and Fertilization*. Victoria, BC: Forest Practices Branch. Available at www.for.gov.bc.ca/hfp/publications/00003/index.htm (accessed April 10, 2009).

Brown, James K., Elizabeth D. Reinhardt, and Kylie A. Kramer. 2003. *Coarse Woody Debris: Managing Benefits and Fire Hazard in the Recovering Forest*. General Technical Report RMRS-GTR-105. Ogden, UT: U.S. Department of Agriculture, Forest Service, Rocky Mountain Research Station.

Chladná, Zuzana. 2007. "Determination of Optimal Rotation Period under Stochastic Wood and Carbon Prices." *Forest Policy and Economics* 9 (8): 1031–45.

Covington, W. Wallace, and Margaret M. Moore. 1994. "Southwestern Ponderosa Pine Forest Structure: Changes Since Euro-American Settlement." *Journal of Forestry* 92 (1): 39–47.

Curtis, Robert O., Dean S. DeBell, Constance A. Harrington, Denis P. Lavender, J. Bradley St. Clair, John C. Tappeiner, and Jack D. Walstad. 1998. *Silviculture for Multiple Objectives in the Douglas-fir Region*. General Technical Report PNW-GTR-435. Portland, OR: U.S. Department of Agriculture, Forest Service, Pacific Northwest Research Station.

Dixon, Gary E. 2002. *Essential FVS: A User's Guide to the Forest Vegetation Simulator*. Internal Report. Fort Collins, CO: U.S. Department of Agriculture, Forest Service, Forest Management Service Center.

Englin, Jeffrey E., Peter C. Boxall, and Grant K. Hauer. 2000. "An Empirical Examination of Optimal Rotations in a Multiple-Use Forest in the Presence of Fire Risk." *Journal of Agricultural and Resource Economics* 25 (1): 14–27.

Englin, Jeffrey E., and John M. Callaway. 1993. "Global Climate Change and Optimal Forest Management." *Natural Resource Modeling* 7 (3): 191–202.

Faustmann, Martin. 1849. "On the Determination of the Value Which Forest Land and Immature Stands Possess for Forestry." In *Martin Faustmann and the Evolution of Discounted Cash Flow*, ed. M. Gane, Institute Paper 42. Oxford: Commonwealth Forestry Institute, Oxford University.

Graham, Russell T., Sarah McCaffrey, and Theresa B. Jain. 2004. *Science Basis for Changing Forest Structure to Modify Wildfire Behavior and Severity*. General Technical Report RMRS-GTR-120. Fort Collins, CO: U.S. Department of Agriculture, Forest Service, Rocky Mountain Research Station.

Haynes, Richard W. 2003. *An Analysis of the Timber Situation in the United States: 1952–2050*. General Technical Report PNW-GTR-560. Portland, OR: U.S. Department of Agriculture, Forest Service, Pacific Northwest Research Station.

Hoen, Hans Fredrik, and Birger Solberg. 1994. "Potential and Economic Efficiency of Carbon Sequestration in Forest Biomass through Silvicultural Management." *Forest Science* 40 (3): 429–51.

Howard, James L. 2003. *U.S. Timber Production, Trade, Consumption, and Price Statistics 1965 to 2002*. Research Paper FPL-RP-615. Madison, WI: U.S. Department of Agriculture, Forest Service, Forest Products Laboratory.

Judd, Kenneth L. 1998. *Numerical Methods in Economics*. Cambridge, MA: MIT Press.

Kindermann, Georg, Michael Obersteiner, Brent Sohngen, Jayant Sathaye, Kenneth Andrasko, Ewald Rametsteiner, Bernhard Schlamadinger, Sven Wunder, and Robert Beach. 2008. "Global Cost Estimates of Reducing Carbon Emissions through Avoided Deforestation." *Proceedings of the National Academy of Sciences* 105 (30): 10302–7.

Martell, David L. 1980. "The Optimal Rotation of a Flammable Forest Stand." *Canadian Journal of Forest Research* 10 (1): 30–34.

McArdle, Richard E., Walter H. Meyer, and Donald Bruce. 1949, 1961. *The Yield of Douglas-fir in the Pacific Northwest*. Technical Bulletin No. 201 (rev.). Washington, DC: U.S. Department of Agriculture, Forest Service.

Miller, Richard E., Gary W. Clendenen, and David Bruce. 1988. *Volume Growth and Response to Thinning and Fertilizing of Douglas-fir Stands in Southwestern Oregon*. General Technical Report PNW-GTR-221. Portland, OR: U.S. Department of Agriculture, Forest Service, Pacific Northwest Research Station.

Miranda, Mario J., and Paul L. Fackler. 2002. *Applied Computational Economics and Finance*. Cambridge, MA: MIT Press.

Morrison, Peter H., and Frederick J. Swanson. 1990. *Fire History and Pattern in a Cascade Range Landscape*. General Technical Report PNW-GTR-254. Portland, OR: U.S. Department of Agriculture, Forest Service, Pacific Northwest Research Station.

Murray, Brian C. 2000. "Carbon Values, Reforestation, and 'Perverse' Incentives under the Kyoto Protocol: An Empirical Analysis." *Mitigation and Adaptation Strategies for Global Change* 5 (3): 271–95.

Omi, Philip N., and Eric J. Martinson. 2002. "Effect of Fuels Treatment on Wildfire Severity." Report submitted to the Joint Fire Science Program Governing Board, Fort Collins, CO. Available at www.fs.fed.us/rm/pubs/rmrs_p029.html (accessed April 10, 2009).

Pohjola, Johanna, and Lauri Valsta. 2007. "Carbon Credits and Management of Scots Pine and Norway Spruce Stands in Finland." *Forest Policy and Economics* 9 (7): 789–98.

Prestemon, Jeffrey P., and Thomas P. Holmes. 2000. "Timber Price Dynamics Following a Natural Catastrophe." *American Journal of Agricultural Economics* 82 (1): 145–60.

Reed, William J. 1984. "The Effects of the Risk of Fire on the Optimal Rotation of a Forest." *Journal of Environmental Economics and Management* 11 (2): 180–90.

———. 1987. "Protecting a Forest against Fire: Optimal Protection Patterns and Harvest Policies." *Natural Resource Modeling* 2 (1): 22–53.

———. 1993. "The Decision to Conserve or Harvest Old-Growth Forest." *Ecological Economics* 8 (1): 45–69.

Reed, W. J., and D. Errico. 1985. "Assessing the Long-Run Yield of a Forest Stand Subject to the Risk of Fire." *Canadian Journal of Forest Research* 15 (4): 680–87.

Reinhardt, Elizabeth and Nicholas L. Crookston, tech. eds. 2003. *The Fire and Fuels Extension to the Forest Vegetation Simulator*. General Technical Report RMRS-GTR-116. Ogden, UT: U.S. Department of Agriculture, Forest Service, Rocky Mountain Research Station.

Richards, Kenneth R., and Carrie Stokes. 2004. "A Review of Forest Carbon Sequestration Cost Studies: A Dozen Years of Research." *Climatic Change* 63 (1–2): 1–48.

Routledge, R. D. 1980. "The Effect of Potential Catastrophic Mortality and Other Unpredictable Events on Optimal Forest Rotation Policy." *Forest Science* 26 (Suppl. to 3): 389–99.

Rummer, Bob, Jeff Prestemon, Dennis May, Pat Miles, John Vissage, Ron McRoberts, Greg Liknes, Wayne D. Shepperd, Dennis Ferguson, William Elliot, Sue Miller, Steve Reutebuch, Jamie Barbour, Jeremy Fried, Bryce Stokes, Edward Bilek, and Ken Skog. 2003. *A Strategic Assessment of Forest Biomass and Fuel Reduction Treatments in Western States*. Washington, DC: U.S. Department of Agriculture, Forest Service. Available at www.fs.fed.us/research/pdf/Western_final.pdf (accessed Apri 10, 2009).

Sedjo, Roger A., and Allen M. Solomon. 1989. "Climate and Forests." In *Greenhouse Warming: Abatement and Adaptation*, ed. Norman J. Rosenburg, William E. Easterling, Pierre R. Crosson, and Joel Darmstadter, 105–19. Washington, DC: Resources for the Future.

Sedjo, Roger A., Joe Wisniewski, Alaric V. Sample, and John D. Kinsman. 1995. "The Economics of Managing Carbon via Forestry: Assessment of Economic Studies." *Environmental and Resource Economics* 6 (2): 139–65.

Smith, James E., Linda S. Heath, and Jennifer C. Jenkins. 2003. *Forest Volume-to-Biomass Models and Estimates of Mass for Live and Standing Dead Trees of U.S. Forests*. General Technical Report NE-298. Newtown Square, PA: U.S. Department of Agriculture, Forest Service, Northeastern Research Station.

Stainback, G. Andrew, and Janaki R. R. Alavalapati. 2004. "Modeling Catastrophic Risk in Economic Analysis of Forest Carbon Sequestration." *Natural Resource Modeling* 17 (3): 299–317.

Stavins, Robert N. 1999. "The Costs of Carbon Sequestration: A Revealed Preference Approach." *American Economic Review* 89 (4): 994–1009.

Stephens, Scott L., and Jason J. Moghaddas. 2005. "Silvicultural and Reserve Impacts on Potential Fire Behavior and Forest Conservation: Twenty-five Years of Experience from Sierra Nevada Mixed Conifer Forests." *Biological Conservation* 125 (3): 369–79.

Stollery, Kenneth R. 2005. "Climate Change and Optimal Rotation in a Flammable Forest." *Natural Resource Modeling* 18 (1): 91–112.

U.S. Environmental Protection Agency (USEPA). 2005. *Greenhouse Gas Mitigation Potential in U.S. Forestry and Agriculture*. EPA-430-R-05-006. Washington, DC: U.S. Environmental Protection Agency, Office of Atmospheric Programs.

van der Werf, Guido R., James T. Randerson, Louis Giglio, G. James Collatz, and Prasad S. Kasibhatla. 2006. "Interannual Variability in Global Biomass Burning Emission from 1997 to 2004." *Atmospheric Chemistry and Physics* 6 (1): 1–19.

van Kooten, G. Cornelis, Clark S. Binkley, and Gregg Delcourt. 1995. "Effect of Carbon Taxes and Subsidies on Optimal Forest Rotation Age and Supply of Carbon Services." *American Journal of Agricultural Economics* 77 (2): 365–74.

van Wagner, C. E. 1977. "Conditions for the Start and Spread of Crown Fire." *Canadian Journal of Forest Research* 7 (1): 23–34.

Westerling, Anthony L., Hugo G. Hidalgo, Daniel R. Cayan, and Thomas W. Swetnam. 2006. "Warming and Earlier Spring Increase Western U.S. Forest Wildfire Activity." *Science* 313 (5789): 940–43.

Wiedinmyer, Christine, and Jason C. Neff. 2007. "Estimates of CO_2 from Fires in the United States: Implications from Carbon Management." *Carbon Balance and Management* 2 (10). Available at www.cbmjournal.com/content/2/1/10.

Winjum, Jack K., Sandra Brown, and Bernhard Schlamadinger. 1998. "Forest Harvests and Wood Products: Sources and Sinks of Atmospheric Carbon Dioxide." *Forest Science* 44 (2): 272–84.

Yoder, Jonathan. 2004. "Playing with Fire: Endogenous Risk in Resource Management." *American Journal of Agricultural Economics* 86 (4): 933–48.